Uncertainty and Resampling Methods

Big Data y Machine Learning para Economía Aplicada

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Agenda

1. Previously ...
2. Uncertainty
3. Resampling methods
4. The Non Parametric Bootstrap
   - The Wild Bootstrap
5. The "XY" Bootstrap
   - Example: Elasticity of Demand for Gasoline
6. How Many Bootstrap Replications?
   - The Bag of Little Bootstraps
7. Recap

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Set up

- We observe (Y,X)
- We choose a predictor $f : \mathcal{X} \to \mathbb{R}$ to minimize expected loss.

$$\hat{f} = \underset{f}{\operatorname{argmin}}\, R(f) \tag{1}$$

$$= \underset{f}{\operatorname{argmin}}\, \mathbb{E}\left[(Y - f(X))^2\right]. \tag{2}$$

Linear model

$$f(X) = X\beta \tag{3}$$

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Uncertainty

Motivation

Analytical: Example 1

- Suppose we have $y_1, y_2, \ldots, y_n$ iid $Y \sim (\mu, \sigma^2)$ (both finite)
- We want to estimate

$$Var(\bar{Y}) \tag{4}$$

Motivation

Analytical: Example 2

- Suppose we have $y_i = \beta X_i + \epsilon_i \; i = 1, \ldots, n \; E(\epsilon|X) = 0 \; V(\epsilon|X) = \sigma^2 I$

$$\hat{\beta} = (X'X)^{-1}X'y \tag{5}$$

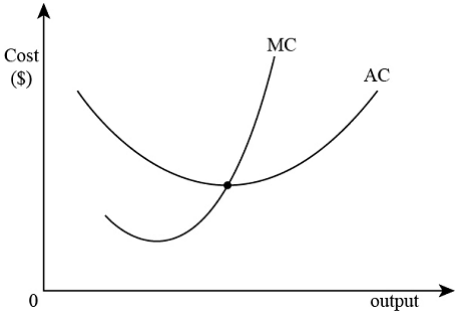
Motivation

Analytical: Example 3

- Suppose we have estimated a cost function of the quadratic form,

$$y_i = \alpha_0 + \alpha_1 x_i + \alpha_2 x_i^2 + \mathbf{z}_i^\top \beta + u_i$$

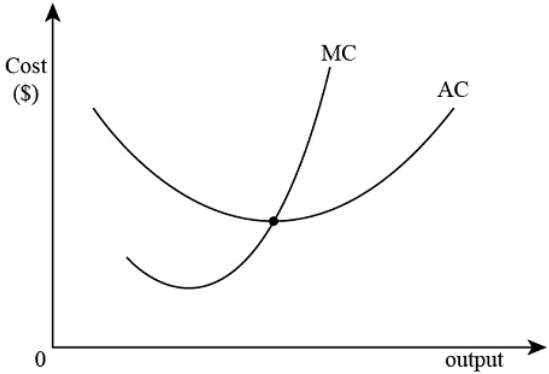
- where
  - y_i is log cost of firm i,
  - $x_i = \log(q_i)$ is log output, and
  - $\mathbf{z}_i$ is a vector of other characteristics of the ith firm.
- I want to minimize average cost.



Motivation

Analytical: Example 3

I want to minimize average cost.



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What are resampling methods?

- Tools that involves repeatedly drawing samples and refitting a model of interest on each sample in order to obtain more information about the fitted model
  - Parameter Assessment: estimate standard errors
  - Model Assessment: finding the best model
  - Model Selection: select the appropriate level of model flexibility

The Bootstrap

- Sometimes the analytical expression of the variance can be quite complicated.
- In these cases bootstrap can be useful
- In German the expression *an den eigenen Haaren aus dem Sumpf zu ziehen* nicely captures the idea of the bootstrap – *"to pull yourself out of the swamp by your own hair."*

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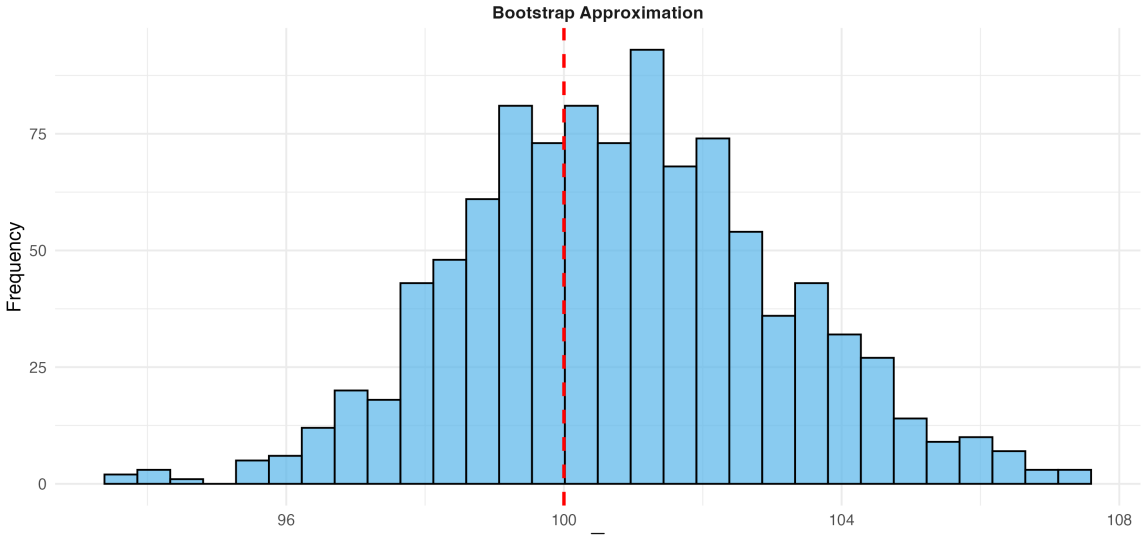
Non Parametric Bootstrap

- Suppose we have $y_1, y_2, \ldots, y_n$ iid $Y \sim F(\mu, \sigma^2)$ (both finite)
- We want to estimate

$$Var(\bar{Y}) \tag{6}$$

- Alternative way (no formula!)
  1. From the n original data points $y_1, y_2, \ldots, y_n$ take a sample *with replacement* of size n
  2. Calculate the sample average of this *"pseudo-sample"* (Bootstrap sample)
  3. Repeat this B times.
  4. Compute the variance

Non Parametric Bootstrap



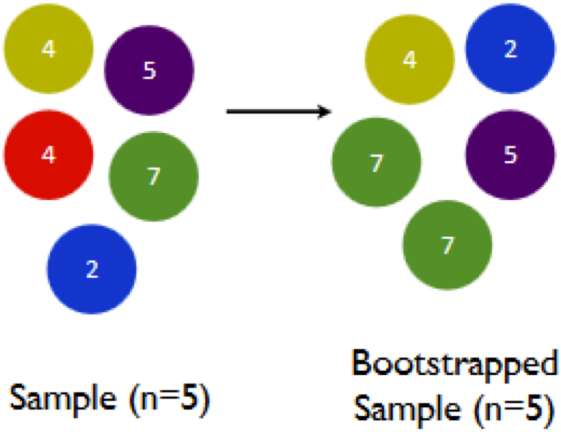
The Non Parametric Bootstrap

Two key properties

- Two key properties of bootstrapping that make this seemingly crazy idea actually work.
  1. Each bootstrap sample must be of the same size (n) as the original sample
  2. Each bootstrap sample must be taken with replacement from the original sample.

The Non Parametric Bootstrap

Sampling with replacement



The Non Parametric Bootstrap

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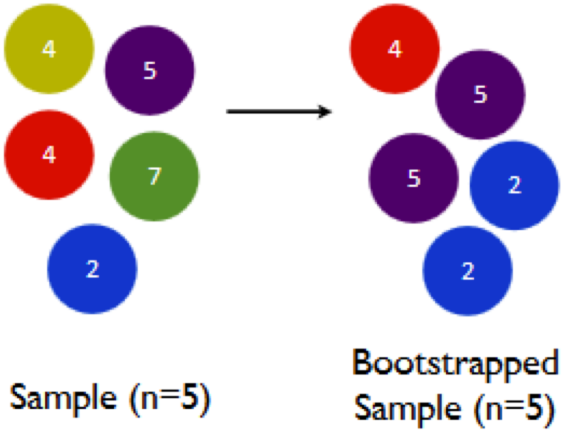

The Non Parametric Bootstrap

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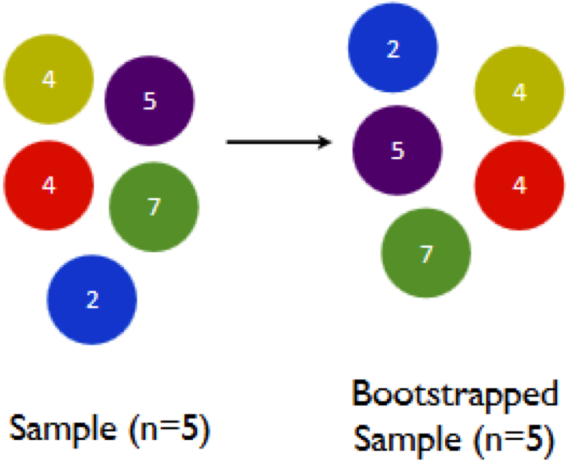
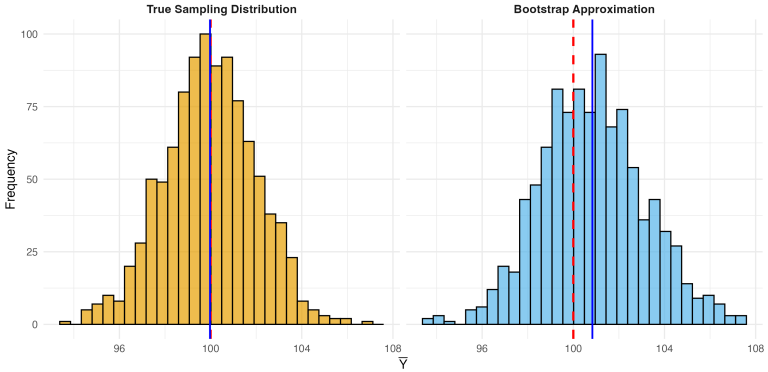


Why bootstrap works?

- The non-parametric bootstrap, relies on empirical data to resample and estimate the sampling distribution.
- This method allows us to understand the variability of a statistic without relying on theoretical distribution assumptions.



Why bootstrap works?

- The key is that the distribution of any estimator or statistic is determined by the distribution of the data.
- While the latter is unknown it can be estimated by the empirical distribution.

Why bootstrap works?

Empirical Distribution Function

- **Empirical distribution function** Given a sample of data $(x_1, \ldots, x_n)$, each an iid realization of some random variable X, we define the empirical cumulative distribution function (ecdf) as:

$$\hat{F}(x) := \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}\{x_i \leq x\} \quad \forall\, x \in \mathbb{R}$$

Why bootstrap works?

Glivenko-Cantelli Theorem

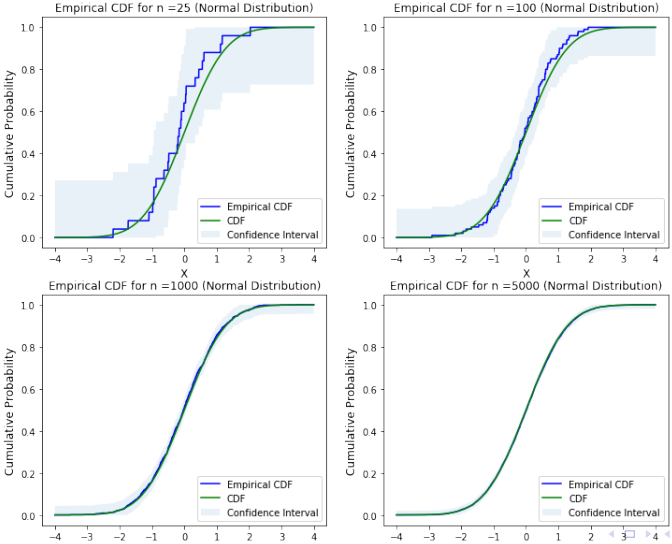
- **Glivenko-Cantelli**: Let $X_1, \ldots, X_n$ be a random sample from a distribution with cdf $F(x)$. Then:

$$\sup_{x \in \mathbb{R}} |\hat{F}(x) - F(x)| \to 0 \quad \text{as } n \to \infty$$

where convergence is in probability.

Why bootstrap works?

Glivenko-Cantelli Theorem



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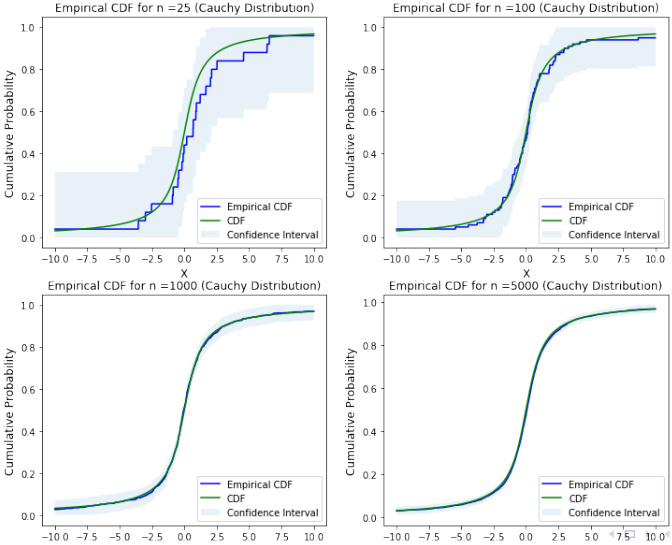

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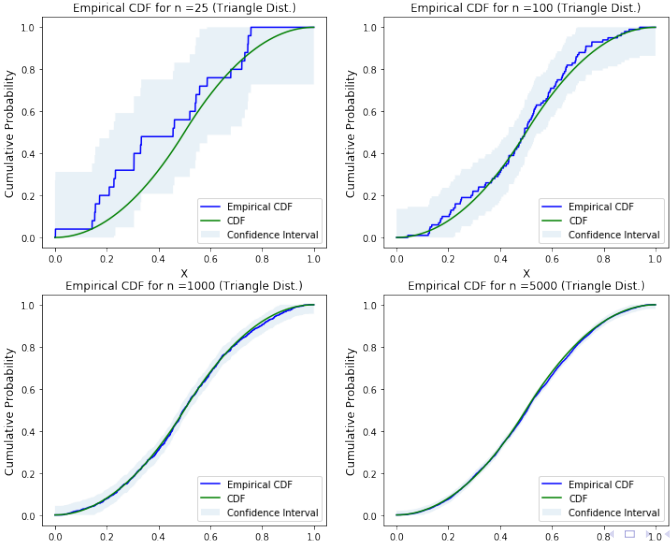

The Non Parametric Bootstrap

Variance in Linear Regression: IID Errors

- $y_i = \alpha_0 + \alpha_1 x_i + \alpha_2 x_i^2 + \mathbf{z}_i^\top \beta + u_i \; i = \{1, \ldots, n\} \; u_i \sim_{iid} F(0, \sigma^2)$

The Non Parametric Bootstrap

Implementation in R

```
fit <- lm(y ~ x)
uhat <- fit$resid
h <- rep(0, R)

for (i in 1:R) {
  yh <- fit$fit + sample(uhat, replace=TRUE)
  b <- lm(yh ~ x)$coef
  h[i] <- exp((1 - b[2]) / (2 * b[3]))
}

var(h)
quantile(h, c(0.025, 0.975))
```

The Wild Bootstrap

Variance in Linear Regression: Heteroskedastic Errors

- For models with heteroskedastic errors, the standard procedure is to use White's heteroskedastic-consistent covariance matrix estimator (HCCME), which can perform poorly in small samples.
- The residual bootstrap can lead to invalid inference by assuming $u_i \mid x_i$ is iid, which may not be the case.
- The wild bootstrap, introduced by Wu (1986) and Liu (1988), refines the bootstrap for heteroskedastic errors.

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The "XY" Bootstrap

- In regression, we need not use the residual bootstrap alone.
- A more direct implementation of the bootstrap would be to resample (x, y) pairs
- This approach is less sensitive to assumptions than the residual-based bootstrap introduced earlier.
  - In particular, it does not assume that the regression errors are iid, so it can accommodate heteroscedasticity, for example.

The "XY" Bootstrap

Implementation in R

```
bhat <- lm(y ~ x)$coef
n <- length(y)
p <- length(bhat)
R <- 500
B <- matrix(0, p, R)

for (i in 1:R) {
  s <- sample(1:n, replace=TRUE)
  B[, i] <- lm(y[s] ~ x[s,])$coef
}

Vhat <- cov(B - bhat)
```

Example: Elasticity of Demand for Gasoline

photo from https://www.dailydot.com/parsec/batman-1966-labels-tumblr-twitter-vine/

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How Many Bootstrap Replications?

- The number of bootstrap replications B is a trade-off between accuracy and computation cost.
- Computation cost is linear in B, while accuracy (e.g., standard errors or p-values) improves as $B^{-1/2}$.
- In empirical research, preliminary calculations can use a modest B, but final results should use a larger B.
- Recommended values: $B = 10,000$ for final calculations, $B = 1,000$ as a minimal choice, and $B = 100$ for rough, quick estimates.

The Bag of Little Bootstraps

- Standard bootstrap resampling can be tedious in large samples, often exceeding storage limits.
- Bickel and Sakov (2002) proposed the m out of n bootstrap method, which involves drawing a smaller sample $m < n$ for each replication.
- This method reduces computation by estimating variability from a smaller sample and rescaling for the reduced size.
- However, choosing m effectively is challenging, and without m being much smaller than n, computational gains are minimal.

Bag of Little Bootstraps

- Kleiner et al. (2014) proposed an alternative scheme for bootstrap resampling that is easily parallelized, improving computational efficiency.
- When n is very large, bootstrap samples can also become very large, even with weighting to reduce the effective sample size.
- Kleiner et al. suggest splitting the sample into G groups, each of size S, with $GS \approx n$.
- Each group undergoes the usual bootstrap on S observations, and the variability measures are averaged across groups to estimate variability for the entire sample.
- Simulations indicate that setting $S = n^{\gamma}$ with $\gamma = 0.7$ works well, significantly speeding up the process.

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Recap

- **Goal:** quantify uncertainty in estimators and predictions.
- **Resampling idea:** repeatedly draw samples and refit the model.
- **Non-parametric bootstrap:** resample from $\hat{F}_n$ to approximate the sampling distribution.
  - **Why it works:** $\hat{F}_n \to F$ (Glivenko–Cantelli).
- **Regression bootstraps:**
  - *Residual bootstrap:* iid errors.
  - *Wild bootstrap:* heteroskedasticity.
  - *XY bootstrap:* resample (x_i, y_i) pairs.
- **Replications:** accuracy improves at rate $B^{-1/2}$ (rule of thumb: $B = 1{,}000$ or $10{,}000$).
- **Big data:** BLB = bootstrap via subsampling + aggregation.